LearnHub

Lecture notes: by date and by subject (minor differences may occur between the two versions)

Class exercises and their solutions

Reading assignments

- Reading assignment 1: Chapter 1 (Assessments) due midnight today.
- Reading assignment 2: Chapter 3-4 (Assessments) due midnight next Monday (17/03/2025).
- You will have one week to work on each reading assignment, every Monday.

Labs:

- Lab 4 material is accessible since yesterday

1

Today

Class exercise on MLE

Probability plots (Section 4.10)

Central limit theorem (Section 4.11)

2

Class exercise

Let $\{X_1, X_2, \ldots, X_5\}$ be a random sample, where X_i's are exponential random variables with an unknown parameter λ with the following PDF:

$$f_X(x;\lambda) = \begin{cases} \lambda e^{-\lambda x} & x > 0, \\ 0 & otherwise. \end{cases}$$

The sample has a mean of 0.6 and a standard deviation of 0.42:

1.3, 0.4, 0.5, 0.6, 0.2

Find the maximum likelihood estimator of λ.

3

Methods of Maximum Likelihood

Suppose that X is a random variable with probability distribution $f(x; \theta)$, where θ is a single unknown parameter. Let $x_1, x_2, \ldots, x_n$ be the observed values in a random sample of size n. Then the **likelihood function** of the sample is

$$L(\theta) = f(x_1; \theta) \cdot f(x_2; \theta) \cdot \cdots \cdot f(x_n; \theta) \qquad (7\text{-}5)$$

Note that the likelihood function is now a function of only the unknown parameter θ. The **maximum likelihood estimator** of θ is the value of θ that maximizes the likelihood function $L(\theta)$.

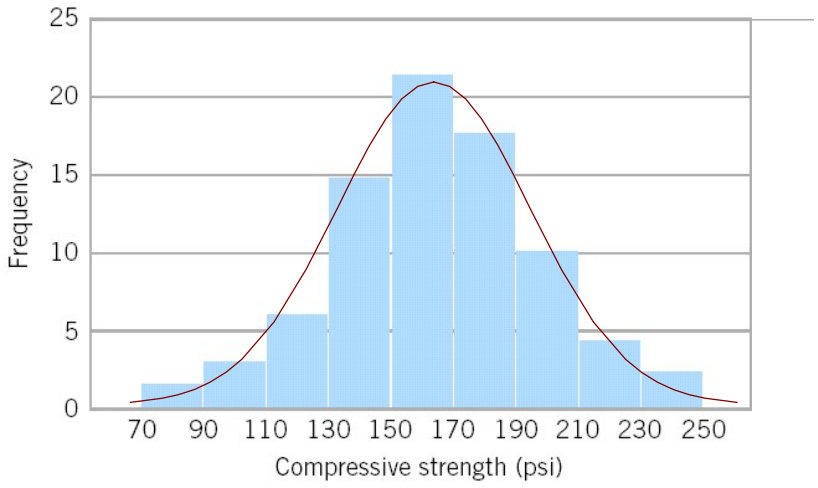
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Probability Plots (Section 4.10)

• Probability plotting is a graphical method for determining whether sample data conform to a hypothesized distribution based on a subjective visual examination of the data.

5

The main idea



6

Quantiles of a distribution versus sample quantiles

Defn: Let *p* be a number between 0 and 1, the *100p*%th **population quantile** of the distribution of a continuous rv *X*, denoted by ς(p), is defined by:

$$p = F(\varsigma(p)) = \int_{-\infty}^{\varsigma(p)} f(y)dy$$

Defn: The 100*k*th **sample quantile** is a data value such that approximately 100*k*% of the observations are at or below this value and approximately 100(1 - *k*)% of them are above it.

- n ordered observations: j^{th} smallest → *100(j-0.5)/n)*th quantile

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A probability plot

A plot of n pairs

*(j*th smallest observation, (*100(j-0.5)/n)*th quantile of the distribution)

on a two-dimensional coordinate system is called a probability plot.

If this plot is a reasonably straight line, then you may conclude that the sample came from the distribution that we used to find quantiles.

We will always use a software to plot these, so we skip the details of computing population and sample quantiles. We will learn in the lab.

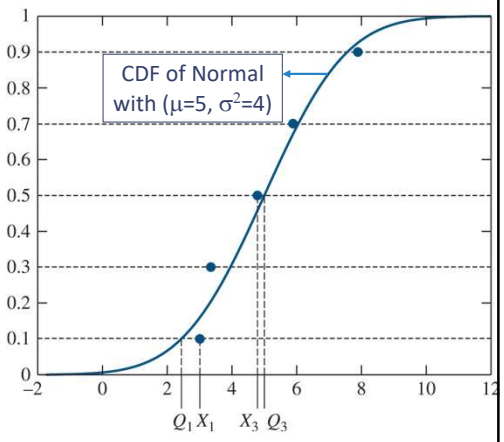
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Example: Normal Data?

3.01, 3.35, 4.79, 5.96, 7.89

$\bar{x} = 5$ and $s = 2$

i	X_i	$(i - 0.5)/5$
1	3.01	0.1
2	3.35	0.3
3	4.79	0.5
4	5.96	0.7
5	7.89	0.9



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$$Z = \frac{X - \mu}{\sigma}$$

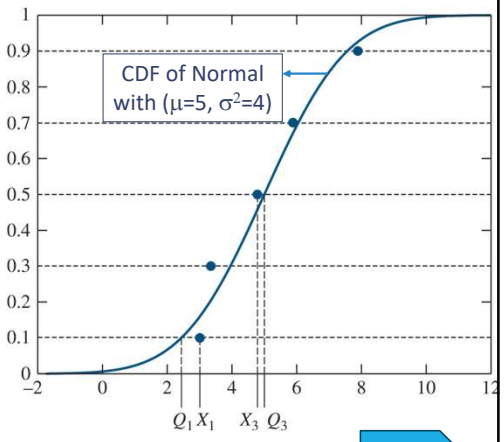
Example: Normal Data?

3.01, 3.35, 4.79, 5.96, 7.89

$\bar{x} = 5$ and $s = 2$

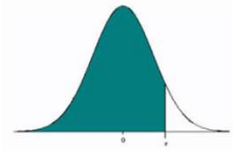
i	X_i	$(i - 0.5)/5$	Q_i
1	3.01	0.1	2.44
2	3.35	0.3	3.95
3	4.79	0.5	5.00
4	5.96	0.7	6.05
5	7.89	0.9	7.56

i	100(i-0.5)/5	zi
1	0.1	-1.28
2	0.3	-0.53
3	0.5	0
4	0.7	0.53
5	0.9	1.28



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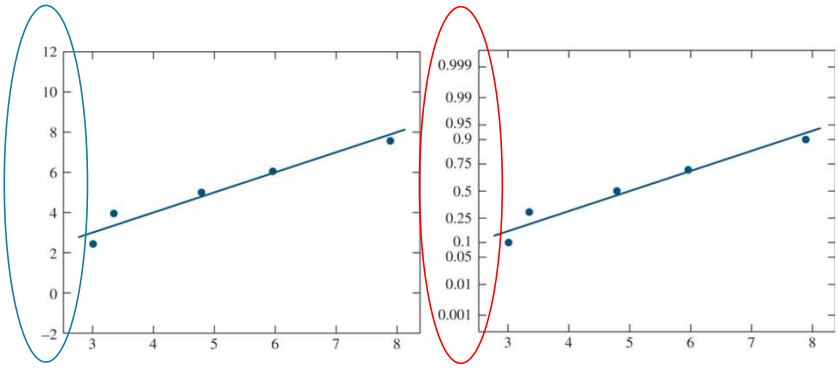
Table of Standard Normal Probabilities for Positive Z-scores



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

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Example

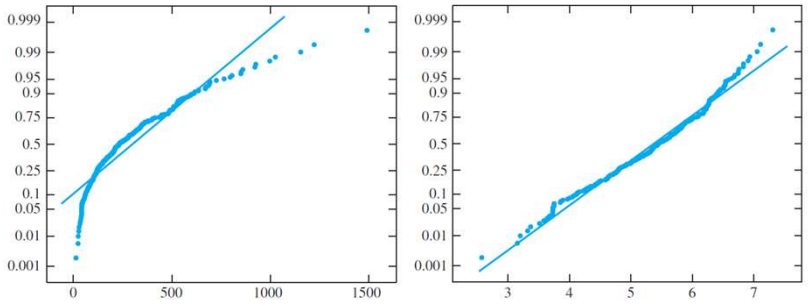


Percentiles of N(5,4)

Percent corresponding to N(5,4)

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Normal Probability Plots

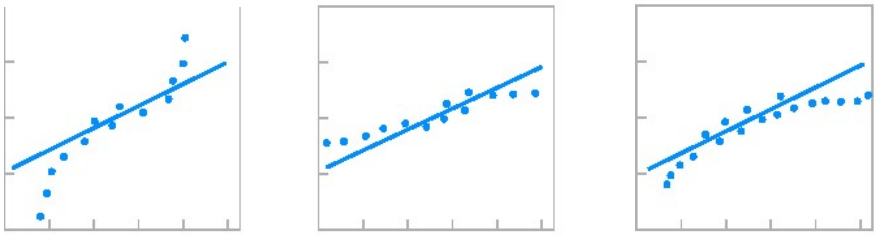


Not normal

Close to normal

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Normal Probability Plots



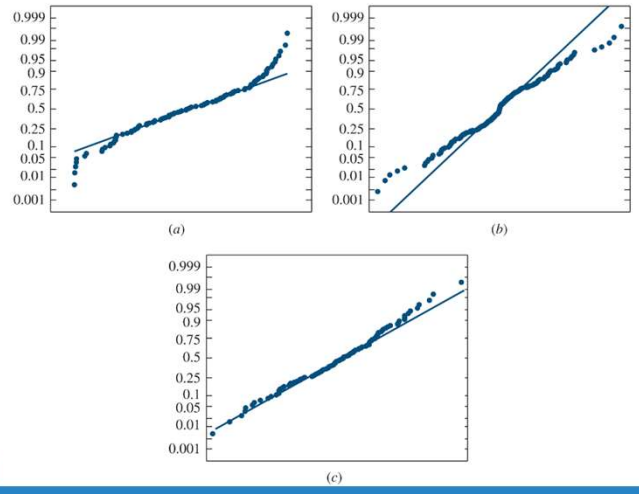
Symmetric and light-tailed distribution

Symmetric and heavy-tailed distribution

A distribution with positive (or right) skew

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Which of these samples appear to be normal?



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$$\Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy$$

Central Limit Theorem (CLT)

$X_1, \ldots, X_n$ be independent identically distributed (iid) rvs with $E(X_i) = \mu$ and $Var(X_i) = \sigma^2$. Then:

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}} = \frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma\sqrt{n}} \rightarrow Z_n \sim N(0,1) \quad \text{as} \quad n \rightarrow \infty,$$

$$P\left(\frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma\sqrt{n}} \leq a\right) \xrightarrow{n\rightarrow\infty} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{a} e^{-\frac{x^2}{2}} dx$$

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Central Limit Theorem

- Central Limit Theorem applet with different distributions.
- You will observe this in Python simulations in the lab exercises and assignments.
- Applet: https://onlinestatbook.com/stat_sim/sampling_dist/